Deriving Black–Scholes as the Limit of the Binomial Model

Claude Sonnet 5

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This note derives the Black–Scholes formula as the $n \to \infty$ limit of the Cox–Ross–Rubinstein (CRR) binomial option pricing model.

Setup

Divide the option's life T into n steps, so each step has length

$$\Delta t = \frac{T}{n}.$$

Using the CRR parameterization:

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}}, \qquad p = \frac{e^{r\Delta t} - d}{u - d},$$

where σ is volatility, r is the risk-free rate, S_0 is today's stock price, K is the strike, and p is the risk-neutral probability of an up move.

After n steps, with j up-moves and $n - j$ down-moves, the stock price is

$$S(j) = S_0 u^j d^{n-j}.$$

The binomial price of a European call is the discounted risk-neutral expected payoff:

$$C_0 = e^{-rT} \sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j} \max\big(S(j) - K,\, 0\big).$$

Step 1: Isolate the in-the-money terms

Let a be the smallest integer j such that $S_0 u^j d^{n-j} > K$. For $j \geq a$ the payoff is simply $S(j) - K$; for $j < a$ it is zero. The sum collapses to

$$C_0 = e^{-rT} \sum_{j=a}^{n} \binom{n}{j} p^j (1-p)^{n-j} \left(S_0 u^j d^{n-j} - K\right),$$

which splits into two sums:

$$C_0 = S_0 e^{-rT} \sum_{j=a}^{n} \binom{n}{j} (pu)^j \big((1-p)d\big)^{n-j} \;-\; K e^{-rT} \sum_{j=a}^{n} \binom{n}{j} p^j (1-p)^{n-j}.$$

The second sum is $P(X \geq a)$ for $X \sim \text{Binomial}(n, p)$. Denote this complementary binomial CDF by

$$\Psi(a; n, p) := \sum_{j=a}^{n} \binom{n}{j} p^j (1-p)^{n-j}.$$

So the second term is $Ke^{-rT}\Psi(a; n, p)$ — already shaped exactly like the $Ke^{-rT}N(d_2)$ term in Black–Scholes.

Step 2: The first sum under a shifted measure

Note the identity, directly from the definition of p:

$$pu + (1-p)d = e^{r\Delta t}.$$

Define a new probability

$$p' := \frac{pu}{pu + (1-p)d} = pu\, e^{-r\Delta t}.$$

Then

$$(pu)^j \big((1-p)d\big)^{n-j} = \big[pu + (1-p)d\big]^n \, p'^j (1-p')^{n-j} = e^{rn\Delta t}\, p'^j (1-p')^{n-j} = e^{rT}\, p'^j (1-p')^{n-j}.$$

Hence

$$\sum_{j=a}^{n} \binom{n}{j} (pu)^j \big((1-p)d\big)^{n-j} = e^{rT}\Psi(a; n, p').$$

Substituting back, the $S_0 e^{-rT}$ prefactor cancels the e^{rT}, giving an *exact* identity valid for every finite n:

$$\boxed{C_0 = S_0\, \Psi(a; n, p') \; - \; Ke^{-rT}\Psi(a; n, p)}$$

This already has the structural shape of Black–Scholes. It remains to show

$$\Psi(a; n, p) \;\longrightarrow\; N(d_2), \qquad \Psi(a; n, p') \;\longrightarrow\; N(d_1) \qquad \text{as } n \to \infty.$$

> *Interpretation:* p' is the risk-neutral probability under the "stock as numeraire" measure — the discrete-time analogue of a Girsanov change of measure. In the Itô's lemma derivation, d_1 and d_2 differ by exactly $\sigma\sqrt{T}$; that shift is precisely this same measure change, expressed here in purely combinatorial terms.

Step 3: Log stock price as a sum of small steps

Under the true risk-neutral measure p, write

$$\ln\left(\frac{S_n}{S_0}\right) = \sum_{i=1}^{n} Y_i, \qquad Y_i = \begin{cases} +\sigma\sqrt{\Delta t} & \text{w.p. } p \\ -\sigma\sqrt{\Delta t} & \text{w.p. } 1-p \end{cases}$$

so that

$$\mathbb{E}[Y_i] = \sigma\sqrt{\Delta t}\,(2p-1), \qquad \text{Var}(Y_i) = \sigma^2 \Delta t \left[1 - (2p-1)^2\right].$$

We need the asymptotic behavior of p as $\Delta t \to 0$.

Step 4: Expand p for small Δt

Taylor-expand each term to first order in Δt:

$$
\begin{aligned}
e^{r\Delta t} &\approx 1 + r\Delta t,\\
u = e^{\sigma\sqrt{\Delta t}} &\approx 1 + \sigma\sqrt{\Delta t} + \tfrac{1}{2}\sigma^2\Delta t + O(\Delta t^{3/2}),\\
d = e^{-\sigma\sqrt{\Delta t}} &\approx 1 - \sigma\sqrt{\Delta t} + \tfrac{1}{2}\sigma^2\Delta t + O(\Delta t^{3/2}).
\end{aligned}
$$

So

$$u - d \approx 2\sigma\sqrt{\Delta t}, \qquad e^{r\Delta t} - d \approx \sigma\sqrt{\Delta t} + \left(r - \tfrac{\sigma^2}{2}\right)\Delta t + O(\Delta t^{3/2}).$$

Dividing,

$$p \approx \frac{1}{2} + \frac{r - \sigma^2/2}{2\sigma}\sqrt{\Delta t}.$$

This is the key asymptotic fact: $p \to 1/2$, with a correction carrying drift $r - \sigma^2/2$ — the same Itô-correction drift that appears in the Black–Scholes lognormal distribution, obtained here purely from a Taylor expansion.

Step 5: Limiting mean and variance

Mean of the sum:

$$n\,\mathbb{E}[Y_i] = n\sigma\sqrt{\Delta t}\,(2p-1) = n\sigma\sqrt{\Delta t}\cdot\frac{(r-\sigma^2/2)\sqrt{\Delta t}}{\sigma} = n\left(r - \tfrac{\sigma^2}{2}\right)\Delta t = \left(r - \tfrac{\sigma^2}{2}\right)T.$$

Variance of the sum: since $(2p-1)^2 = O(\Delta t)$ is negligible next to 1,

$$n\,\mathrm{Var}(Y_i) \approx n\sigma^2\Delta t = \sigma^2 T.$$

By the Central Limit Theorem (De Moivre–Laplace),

$$\ln\!\left(\frac{S_n}{S_0}\right) \xrightarrow{d} \mathcal{N}\!\left(\left(r - \tfrac{\sigma^2}{2}\right)T,\ \sigma^2 T\right) \qquad \text{as } n \to \infty.$$

This is exactly the risk-neutral lognormal distribution Black–Scholes assumes — derived here, not assumed.

Step 6: $\Psi(a; n, p) \to N(d_2)$

The event $j \geq a$ is the event $S_n > K$. Standardizing the limiting normal distribution from Step 5,

$$\Psi(a; n, p) \;\longrightarrow\; N(d_2), \qquad d_2 = \frac{\ln(S_0/K) + \left(r - \sigma^2/2\right)T}{\sigma\sqrt{T}}.$$

Step 7: $\Psi(a; n, p') \to N(d_1)$

Repeating Step 4's expansion for p' instead of p gives the same form with the opposite sign on the volatility term:

$$p' \approx \frac{1}{2} + \frac{r + \sigma^2/2}{2\sigma}\sqrt{\Delta t}.$$

Repeating Steps 5–6 with p' gives

$$\Psi(a; n, p') \;\longrightarrow\; N(d_1), \qquad d_1 = \frac{\ln(S_0/K) + \left(r + \sigma^2/2\right)T}{\sigma\sqrt{T}} = d_2 + \sigma\sqrt{T}.$$

Conclusion

Taking $n \to \infty$ in the exact identity from Step 2,

$$C_0 = S_0\,\Psi(a; n, p') - Ke^{-rT}\,\Psi(a; n, p),$$

yields the Black–Scholes formula:

$$\boxed{C = S_0\,N(d_1) \;-\; Ke^{-rT}N(d_2)}$$

What this proof teaches

1. **Where d_1 and d_2 come from.** In the PDE derivation they emerge from solving a differential equation. Here they are standardized counts of "how many up-moves are needed to finish in the money," under two different probability measures. The gap $\sigma\sqrt{T}$ between them is the discrete footprint of a measure change — Girsanov's theorem, built from elementary combinatorics.

2. **Where the $-\sigma^2/2$ drift correction comes from.** In the Itô's lemma derivation this term comes from the second-order (quadratic variation) term in Itô's formula. Here it falls out of a plain Taylor expansion of p. Both routes encode the same fact — averaging a multiplicative (lognormal) process pulls the median below the mean — via entirely different machinery.

3. **The whole proof rests on the Central Limit Theorem.** Black–Scholes pricing is fundamentally a CLT statement about a discrete random walk. The continuous-time stochastic calculus version is more powerful and generalizes further, but the essential content — many small independent bets summing to a lognormal outcome — is 18th/19th century probability theory (De Moivre–Laplace).